

Electric field.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

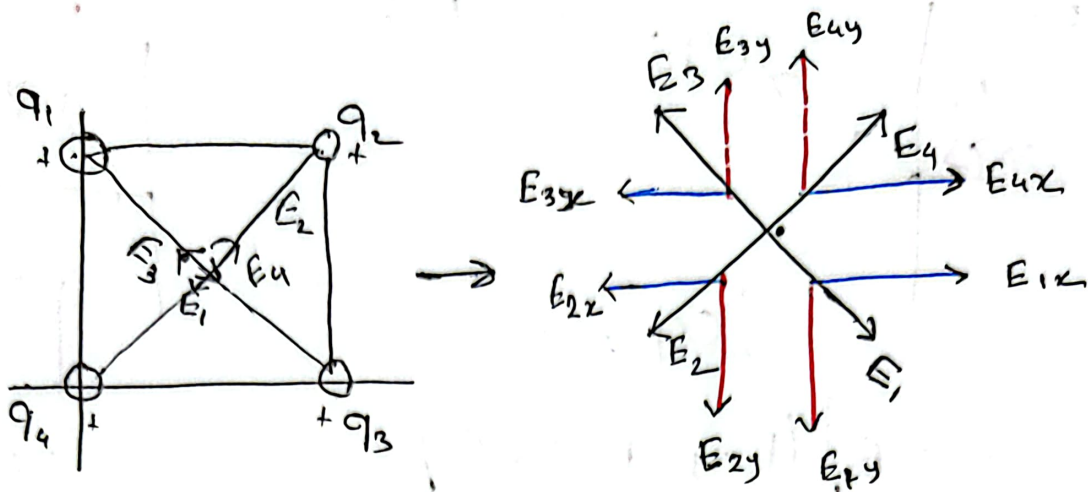
* Electric Force, $F_{es} = k \frac{q_1 q_2}{r^2}$

$$E = \frac{F}{q} = \frac{k \frac{q_1 q_2}{r^2}}{q} = \frac{kq}{r^2}$$

$$F = mg$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$E_{net} = E_1 + E_2$$



$$E_{xnet} = E_{1x} + E_{4x} - E_{2x} - E_{3x}$$

$$E_{ynet} = E_{4y} + E_{3y} - E_{1y} - E_{2y}$$

$$E_{net} = \sqrt{E_{xnet}^2 + E_{ynet}^2}$$

$$\theta = \tan^{-1} \left(\frac{E_{ynet}}{E_{xnet}} \right) \quad [\text{K or rad}] \rightarrow \text{Ans}$$

$$E_{net} = E_1 - E_2$$

$$= \frac{kq_1}{x^2} - \frac{kq_2}{(n-x)^2}$$

$$\frac{q_1}{x^2} = \frac{q_2}{(n-x)^2}$$

$$\frac{5}{x^2} = \frac{2}{(2-x)^2}$$

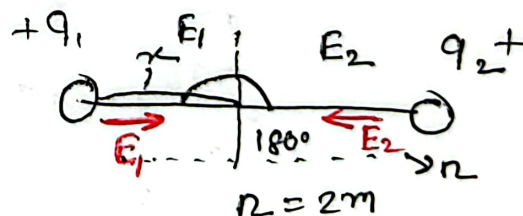
$$\frac{5}{x^2} = \frac{2}{4-4x+x^2}$$

$$5(4-4x+x^2) = 2x^2$$

$$\Rightarrow 3x^2 - 20x + 20 = 0$$

$$x = 5.44 \text{ or } 1.22$$

Ans

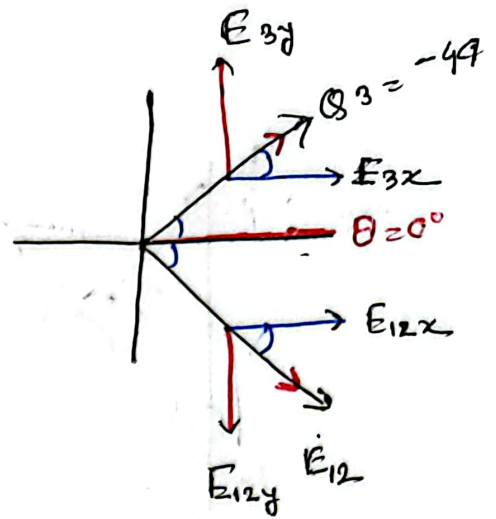
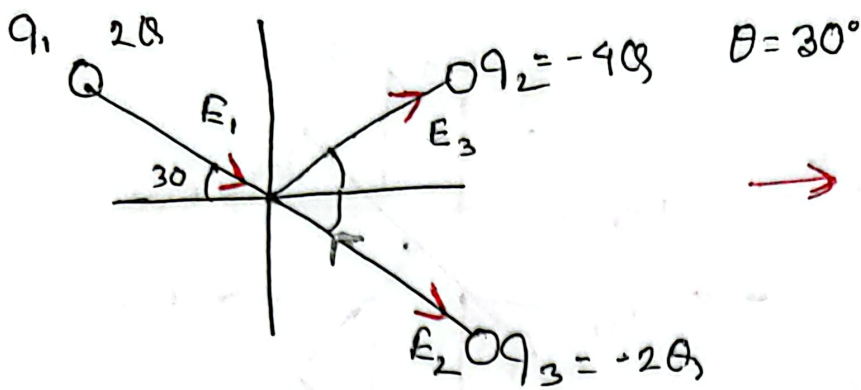


$$q_1 = 5 \text{ C}$$

$$q_2 = 2 \text{ C}$$

$$E_{net} = 0? \quad (\text{correct})$$

$$x = ?$$



$$E_{1x} = E_1 \cos \theta \quad E_{3y} = E_3 \sin \theta$$

$$\vec{E}_{12} = \vec{E}_1 + \vec{E}_2$$

$$q_{13} = 49$$

$$= \frac{k}{d^2} (q_1 + q_2)$$

$$= \frac{49k}{d^2}$$

$$E_{12x} = E_{12} \cos \theta = \frac{49k}{d^2} \cos \theta$$

$$E_{3x} = E_{3x} \cos \theta = \frac{49k}{d^2} \cos \theta$$

$$\therefore E_{xnet} = \frac{49k}{d^2} \cos \theta + \frac{49k}{d^2} \cos \theta$$

$$= \frac{89k}{d^2} \cdot 2 \cos \theta$$

$$= 2 \left(\frac{49k}{d^2} \cos 30^\circ \right)$$

$$\therefore E_{net} = \sqrt{E_{xnet}^2 + 0^2}$$

$$= \frac{89k}{d^2} \cos 30^\circ$$

$$\tan \theta = \frac{E_{ynet}}{E_{xnet}}$$

$$\theta = \tan^{-1} \left(\frac{0}{\dots} \right)$$

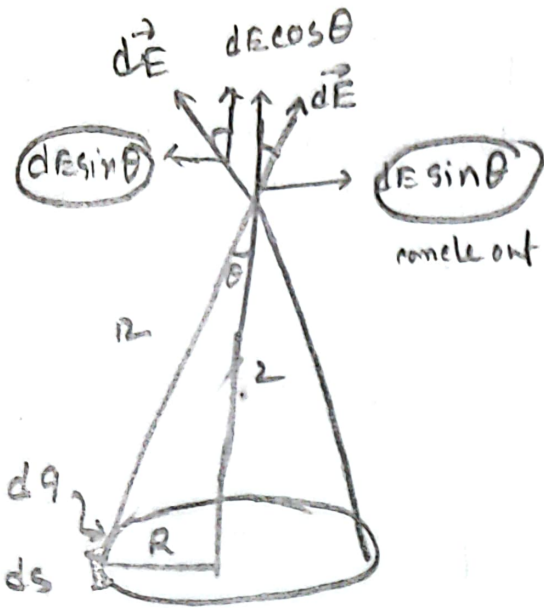
$$= 0^\circ$$

$$E_{ynet} = -E_{12y} \sin \theta + E_{3y} \sin \theta$$

$$= -\frac{49k}{d^2} \sin 30^\circ + \frac{49k}{d^2} \sin 30^\circ$$

$$= 0$$

Electric Field due to the line charge.



$$r^2 = R^2 + z^2$$

$$r = \sqrt{R^2 + z^2}$$

$$\cos \theta = \frac{z}{r}$$

$$= \frac{z}{\sqrt{R^2 + z^2}}$$

Per unit length, charge = λ
 10 unit " " " = 10 λ

$$dq = \lambda ds$$

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda ds}{(\sqrt{R^2 + z^2})^2}$$

$$\therefore dE \cos \theta = k \cdot \frac{\lambda ds}{(\sqrt{R^2 + z^2})^2} \cos \theta$$

$$\therefore dE \cos \theta = \frac{k \lambda ds}{\sqrt{(R^2 + z^2)^2}} \cdot \frac{z}{\sqrt{R^2 + z^2}}$$

$$= \frac{k \lambda ds z}{(R^2 + z^2)^{3/2}}$$

E_F line charge

$$E = \frac{qz}{4\pi\epsilon_0 (R^2 + z^2)^{3/2}}$$

$z \gg R$

$$\hookrightarrow (R^2 + z^2) \rightarrow z^2$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{z^2}$$

Electric Dipole

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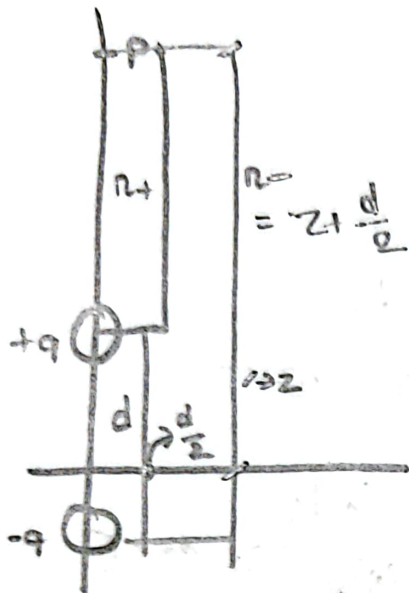
$$E = \frac{1}{2\pi\epsilon_0} \cdot \frac{qd}{z^3}$$

$$|P| = qd$$

↳ magnitude of Electric dipole moment

$$\begin{aligned} z &\gg d \\ z &\gg \frac{d^2}{4} \end{aligned}$$

$$E = E_+ - E_{(-)}$$



P direction $\ominus \rightarrow \oplus$

$z \rightarrow$ distance between ed.m ($\vec{p}$) and mid point of the dipole

Torque

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$F = qE$$

$$\tau = pE \sin \theta$$

$$p = qd$$

$$= Fd \sin \theta$$

$$\begin{aligned} \text{max Torque} &= pE \sin 90 \\ &= pE \end{aligned}$$

Dipole on electric field.

$$\text{work done ; } dW = \tau d\theta$$

Electric field on charge disk

$$U = -W$$

$$U = -pE \cos \theta$$

$$U = -\vec{p} \cdot \vec{E}$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

↳ potential energy of a dipole

Work done on the dipole by electric field

$$W = -\Delta U$$

$$= -(U_f - U_i)$$

↓
external agent $W = (U_f - U_i)$